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TAC 249 Final Project Part 1

Part 1

1.

```
SELECT r.region, AVG(s.percent_qualified) AS avg_percent
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
GROUP BY r.region
ORDER BY avg_percent DESC
LIMIT 1;
```

region	avg_percent
West	88.2754550000

This query joins states to their regions and calculates the average percentage of buildings suitable for solar installation in each region. It then returns the region with the highest average value.

2.

```
SELECT
  ROUND(SUM(yearly_sunlight_kwh_n) / 10000000, 2) AS north_sum,
  ROUND(SUM(yearly_sunlight_kwh_s) / 10000000, 2) AS south_sum,
  ROUND(SUM(yearly_sunlight_kwh_e) / 10000000, 2) AS east_sum,
  ROUND(SUM(yearly_sunlight_kwh_w) / 10000000, 2) AS west_sum,
  ROUND(SUM(yearly_sunlight_kwh_f) / 10000000, 2) AS flat_sum
FROM project_sunroof_state;
```

north_sum	south_sum	east_sum	west_sum	flat_sum
7413.12	33034.76	19534.03	23914.91	63379.96

This query adds up yearly sunlight values for each roof orientation across all states. Each total is divided by 10 million so the results are easier to read and compare.

3.

```
SELECT s.state_name, s.number_of_panels_total
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'West'
ORDER BY s.number_of_panels_total DESC
LIMIT 3;
```

state_name	number_of_panels_to...
California	655000000
Arizona	147000000
Washington	108000000

This query filters the data to states in the West region and sorts them by the total number of solar panels. It returns the top three West states with the highest number of panels.

4.

```
SELECT r.region, COUNT(*) AS state_count
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND s.existing_installs_count > 1000
GROUP BY r.region;
```

region	state_cou...
West	9
Northeast	5
South	9
Midwest	6

This query counts how many states in each region have more than 1,000 buildings estimated to have solar installations. The results show how solar adoption varies by region.

5.

```
SELECT r.region,
       SUM(s.existing_installs_count) / SUM(s.count_qualified) AS install_ratio
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
```

```
GROUP BY r.region
ORDER BY install_ratio ASC
LIMIT 1;
```

region	install_rat...
Midwest	0.0014

This query calculates, for each region, the ratio of installed solar buildings to suitable buildings. It then returns the region with the lowest installation ratio.

6.

```
SELECT s.state_name AS state, s.carbon_offset_metric_tons
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'Northeast'
  AND s.carbon_offset_metric_tons >
      (SELECT AVG(s2.carbon_offset_metric_tons)
       FROM project_sunroof_state s2, region_lookup r2
       WHERE s2.state_name = r2.state
         AND r2.region = 'Northeast');
```

state	carbon_offset_metric_t...
New Jersey	21400000.000
New York	21900000.000
Pennsylvania	25700000.000

This query finds the average carbon offset for the Northeastern region and returns only those Northeastern states whose carbon offset is higher than that average.

7.

```
SELECT r.region, s.state_name AS state
```

```
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'Northeast'
ORDER BY s.carbon_offset_metric_tons DESC
LIMIT 3;
```

region	state
Northeast	Pennsylvania
Northeast	New York
Northeast	New Jersey

This query looks only at Northeastern states and ranks them by carbon offset from highest to lowest. It returns the top three states with the best carbon offset performance.

8.

```
SELECT direction, SUM(yearly_sunlight) AS yearly_sunlight
FROM (
  SELECT 'north' AS direction, yearly_sunlight_kwh_n AS yearly_sunlight
  FROM project_sunroof_state
  UNION ALL
  SELECT 'south', yearly_sunlight_kwh_s
  FROM project_sunroof_state
  UNION ALL
  SELECT 'east', yearly_sunlight_kwh_e
  FROM project_sunroof_state
  UNION ALL
  SELECT 'west', yearly_sunlight_kwh_w
  FROM project_sunroof_state
) t
GROUP BY direction
ORDER BY yearly_sunlight DESC
LIMIT 1;
```

direction	yearly_sunlight
south	330347566911.703

This query combines sunlight values from the north, south, east, and west roof orientations and sums them across all states. It then returns the direction with the greatest total yearly sunlight.

9.

```
SELECT 'max_qualified' AS attribute,
      ROUND(MAX(s.percent_qualified), 2) AS percent_qual
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'West'

UNION ALL

SELECT 'avg_qualified',
      ROUND(AVG(s.percent_qualified), 2)
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'West'

UNION ALL

SELECT 'min_qualified',
      ROUND(MIN(s.percent_qualified), 2)
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'West';
```

attribute	percent_qual
max_qualified	96.43
avg_qualified	88.28
min_qualified	75.68

This query calculates the maximum, average, and minimum percentage of qualified buildings for states in the West region. The results are presented together in a single table.

10.

```
SELECT LEFT(state_name, 1) AS initial,
      COUNT(*) AS numStates
FROM project_sunroof_state
GROUP BY initial
HAVING COUNT(*) >= 5
ORDER BY initial;
```

initial	numStates
A	6
C	5
M	9
N	8
T	5

This query groups states by the first letter of their name and counts how many states fall under each letter. Only letters with five or more states are included in the output.

11.

```
SELECT state_name,
      (yearly_sunlight_kwh_s +
       yearly_sunlight_kwh_e +
       yearly_sunlight_kwh_w +
       yearly_sunlight_kwh_f) AS yearlyKw
FROM project_sunroof_state
ORDER BY yearlyKw DESC
LIMIT 1;
```

state_name	yearlyKw
California	230300000000.000

This query calculates total yearly sunlight for each state using all roof orientations except north. It then returns the state with the highest total sunlight value.

12.

```
SELECT r.region,
      (SUM(s.existing_installs_count) / SUM(s.count_qualified)) * 100 AS install_percentage
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
GROUP BY r.region;
```

region	install_percenta...
South	0.7016
West	2.8618
Northeast	0.7082
Midwest	0.1398

This query calculates the percentage of buildings with solar installations out of all suitable buildings for each region. The results show regional differences in solar adoption rates.

13.

```
SELECT r.region, s.state_name, s.percent_qualified
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'South'
  AND s.percent_qualified =
      (SELECT MAX(s2.percent_qualified)
       FROM project_sunroof_state s2, region_lookup r2
       WHERE s2.state_name = r2.state
         AND r2.region = 'South')

UNION ALL

SELECT r.region, s.state_name, s.percent_qualified
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'West'
  AND s.percent_qualified =
```

```
    (SELECT MAX(s2.percent_qualified)
     FROM project_sunroof_state s2, region_lookup r2
     WHERE s2.state_name = r2.state
       AND r2.region = 'West')

UNION ALL

SELECT r.region, s.state_name, s.percent_qualified
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'Northeast'
  AND s.percent_qualified =
    (SELECT MAX(s2.percent_qualified)
     FROM project_sunroof_state s2, region_lookup r2
     WHERE s2.state_name = r2.state
       AND r2.region = 'Northeast')

UNION ALL

SELECT r.region, s.state_name, s.percent_qualified
FROM project_sunroof_state s, region_lookup r
WHERE s.state_name = r.state
  AND r.region = 'Midwest'
  AND s.percent_qualified =
    (SELECT MAX(s2.percent_qualified)
     FROM project_sunroof_state s2, region_lookup r2
     WHERE s2.state_name = r2.state
       AND r2.region = 'Midwest');
```

region	state_name	percent_qualifi...
South	Oklahoma	92.484939
West	Hawaii	96.429936
Northeast	New York	80.032466
Midwest	Nebraska	84.731339

This query identifies the state in each region with the highest percentage of buildings suitable for solar installation. Each region's top state is returned in a single combined result.

14.

```
SELECT state_name,
    (count_qualified - existing_installs_count) AS diff
FROM project_sunroof_state
ORDER BY diff DESC
LIMIT 1;
```

state_name	diff
California	8286070

This query measures untapped solar potential by subtracting the number of installed buildings from the number of suitable buildings. It returns the state with the largest difference.

15.

```
SELECT state_name,
    carbon_offset_metric_tons / existing_installs_count AS offset_per_building
FROM project_sunroof_state
WHERE existing_installs_count > 0
ORDER BY offset_per_building DESC
LIMIT 1;
```

state_name	offset_per_build...
South Dakota	49903.5140952

This query calculates the average carbon offset per building with solar installation for each state. It returns the state with the highest carbon offset per installed building.